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SHELF SPACE ALLOCATION IN RETAILING: A LITERATURE REVIEW

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Abstract. Efficient allocation of shelf space is vital for attracting customers and maximizing profits in the retail industry, particularly given limited display areas. This paper offers a comprehensive review of shelf-space allocation (SSA) modeling, optimization techniques, and relevant case studies from 1969 to 2023. We categorize the literature into five key areas: first, mathematical optimization, which includes deterministic, uncertain, dynamic, and joint optimization models that form the foundation of SSA literature and enhance decision-making in complex retail environments. Second, we explore data mining techniques, demonstrating how retailers can implement consumer preference insights and purchasing patterns to transition from intuition-based decisions to data-driven strategies. Third, the case studies section illustrates real-world applications of SSA, highlighting the challenges and successes faced by retailers. Fourth, we synthesize methodologies, evaluating various SSA approaches through empirical studies that identify best practices and guide efficient resource utilization. Finally, we outline significant gaps in current research and suggest directions for future inquiry, encouraging ongoing exploration of innovative methods to improve shelf-space strategies. By systematically addressing these categories, this paper aims to provide a clearer understanding of the complexities of SSA, ultimately contributing to both academic discourse and practical applications within the retail sector.

Mathematics Subject Classification. 90B05, 90B80, 90C90.

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1. Introduction

Shelf space is prominent as an expensive and limited resource to attract more consumers. A large number of products compete with each other to obtain more shelf space. Thus, a key short-term decision for retailers is to make a portfolio of product brands and determine the allocation of display space for each brand [51]. The objective of shelf space allocation (SSA) can be considered as the improvement of retail performance measured by indicators such as average expenditure per shopper or turnover per square foot Buttle [18]. SSA benefits can be summarized as decreasing inventory, increasing sales, and more customer satisfaction [100]. The decision of SSA can then be considered a critical issue in retail operations management.

Supply chain management and retailing strategies are greatly involved with the type of delivering products [25]. Generally, three types of products can be introduced in retailing literature considering the space elasticity, which is defined as the ratio of additional product sales to additional space allocated [34]: (1) Unresponsive prod-

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ucts (commodities) such as sugar, salt, and spices. These products are quite price inelastic because consumers usually buy them when needed and therefore changes in space allocated will lead to little reaction; (2) General use products (staples) such as breakfast foods; and (3) Occasional purchase products (impulse buys) such as candies which are unplanned and are highly space-elastic. Elasticity increases as moving from commodities to general-use products and impulse buys [31].

To the best of our knowledge, the state-of-the-art review by Hübner and Kuhn [57] is the only review paper related to SSA. However, their study just addressed the challenges of retail category planning by analyzing and comparing decision support systems for assortment and shelf space management. They highlighted the limitations of existing quantitative models and commercial software, which often overlook critical factors such as space-elastic demand and substitution effects. They only revealed that current planning models fail to account for the complexities of limited shelf space and the impacts of product delisting or stockouts on consumer choices. In the current research, we present an updated review that specifically focuses on SSA and also covers other research methodologies required for both researchers and managers. The paper proceeds as follows: Research methodology and data statistics including publication trend as well as top contribution authors and journals presented in Section 2. Section 3 provides a classification of shelf-space allocation. Moreover, a basic SSA model and an overview of the research on each class are studied in this section. Finally, the paper closes in Section 4 with a discussion of important gaps in that framework and suggestions for some future research directions.

2. Research methodology and data statistics

To identify relevant articles in the literature, the study was conducted using Web of Science, SCOPUS, Google scholar, Taylor & Francis, Emerald and Wiley to search papers between 1969 and 2023 covering published journal papers in scientific English language journals. Using keywords "shelf space allocation", "display space allocation", "retail shelf allocation", "retail space allocation", and "shelf space optimization", 94 journal articles were found including case studies, conceptual modeling, surveys and mathematical modeling. Our investigation shows that the articles found under this set of keywords are the most relevant works to the SSA research area. We excluded conference papers, books, theses, dissertations, and unpublished papers. Including books and conference papers could result in a larger number of records. The framework of inclusion and exclusion of researches, is highlighted in the Figure 1. Each article was reviewed and analyzed in detail to develop a classification framework. Further, each article was classified according to the year of publication and type of journal.

In the past decades, several models have been introduced dealing with shelf space allocation and product assortment [21, 41, 55, 61, 62]. One can divide SSA literature into two subsets, where the first one focuses on testing the elasticity of shelf space to see whether shelf space has any effect on sales and profitability or not (see [28, 34, 60, 70, 101]). The second subset tries to provide optimal SSA policies which will be discussed in detail later (see [83, 85, 106]).

Figure 2 shows the trend of articles published in SSA literature during the last decades. It illustrates the trend of articles published in SSA literature over the past decades, revealing a nearly linear growth in publications.

This increase can be attributed to several factors: as competition in retail environments intensifies, effective shelf-space allocation has become critical for retailers aiming to optimize product visibility and maximize sales, a heightened interest in understanding consumer preferences and shopping behavior has driven research in SSA, as retailers seek to align their product placement strategies with consumer demand, the intersection of disciplines such as operations research, marketing, and supply chain management and development of more complex analytical tools and mathematical models has facilitated the exploration of complex SSA problems, encouraging researchers to publish their findings. This growth in SSA research reflects its significance within the broader context of retail and supply chain dynamics.

Table 1 shows the journals in which SSA papers appeared. It shows an increasing rate of published articles, noticeably after 2000. The increase in published articles on shelf-space allocation (SSA) in journals after 2000 can be attributed to several factors. Firstly, the growing complexity and competitiveness of retail environments have

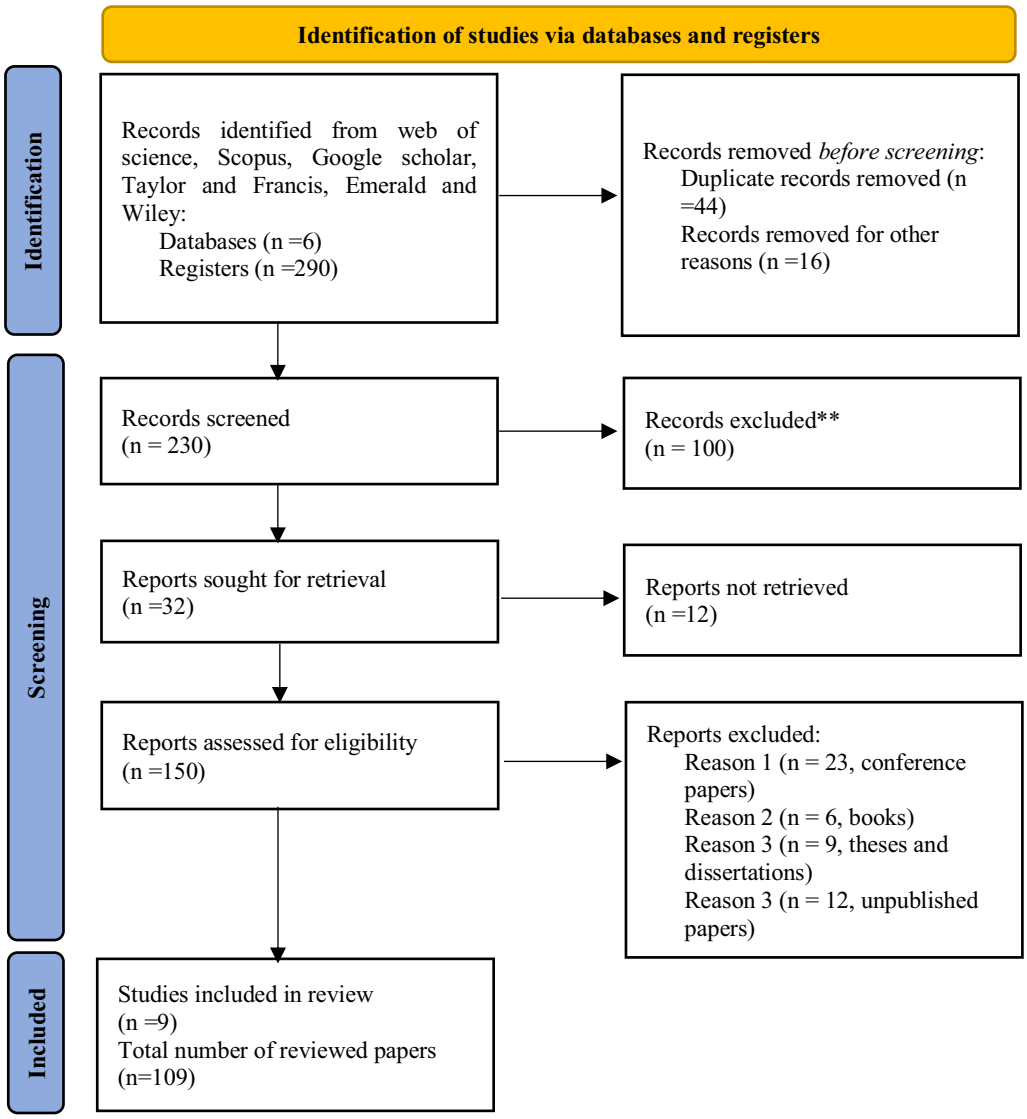

FIGURE 1. The framework of the inclusion and exclusion of reviewed researches.

increased the importance of effective shelf-space management as a critical component of supply chain efficiency and marketing strategy. As retailers seek to optimize their product placements to maximize sales and enhance customer experience, the demand for research in this area has intensified. Additionally, many of the journals listed in Table 1 focus on operations management, logistics, and marketing – disciplines that are increasingly recognizing the strategic significance of shelf-space allocation. After 2000, advancements in technology, such as data analytics and modeling techniques, have also facilitated more sophisticated SSA research. Consequently, the journals highlighted in Table 1 have become key platforms for disseminating innovative findings and practical insights in the SSA domain. The statistics show that more than 40 journals have contributed to publishing 94 papers. Moreover, it is found that the following journals have published 50 articles, representing almost 71% of papers published.

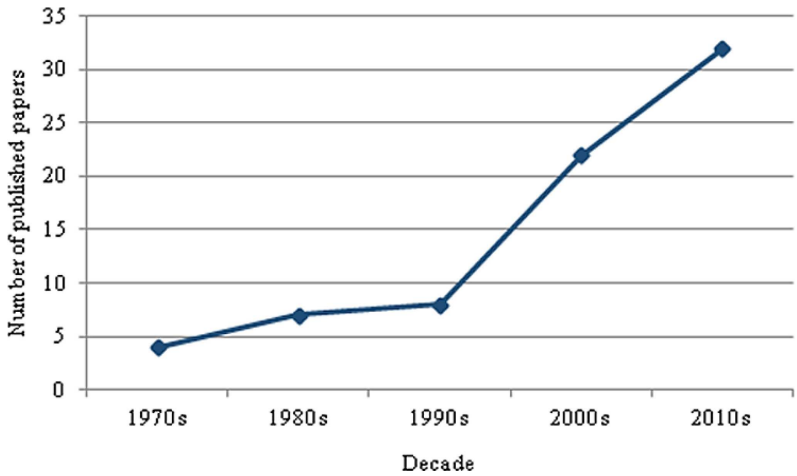

FIGURE 2. Trend of publishing in the area of shelf space allocation.

TABLE 1. The publishing journals contributing to the area of shelf space allocation.

Source	Publication decade						
	1970s	1980s	1990s	2000s	2010s	2020s	Total
European Journal of Operational Research	1			4	7	3	15
Journal of Retailing		1	2	2	3	1	9
Journal of the Operational Research Society		3		2	3		8
Production and Operations Management					5	1	6
International Journal of Production Economics			1	1	3		5
Expert Systems with Applications				1	2	1	4
Computers & Industrial Engineering				1	1	1	3
Management Science		1		1			2
Annals of Operations Research					2		2
Decision Sciences			1	1			2
Decision Support Systems				1	1		2
Others	4	2	3	9	17	16	51
Total	5	7	7	23	44	23	109

3. Proposed SSA classifications

At the highest level, the SSA literature can be classified as illustrated in Figure 3. It should be mentioned that some papers may belong to more than one category. The rationale behind the division into five categories, each serving a distinct purpose within the field.

Mathematical optimization

This category represents the foundation of SSA literature, where quantitative methods are employed to derive optimal solutions. The subcategories reflect different problem settings:

– **Deterministic models:** these models assume known parameters, providing clear, actionable strategies. They are essential in scenarios where data is reliable and predictable, serving as a benchmark for more complex models.

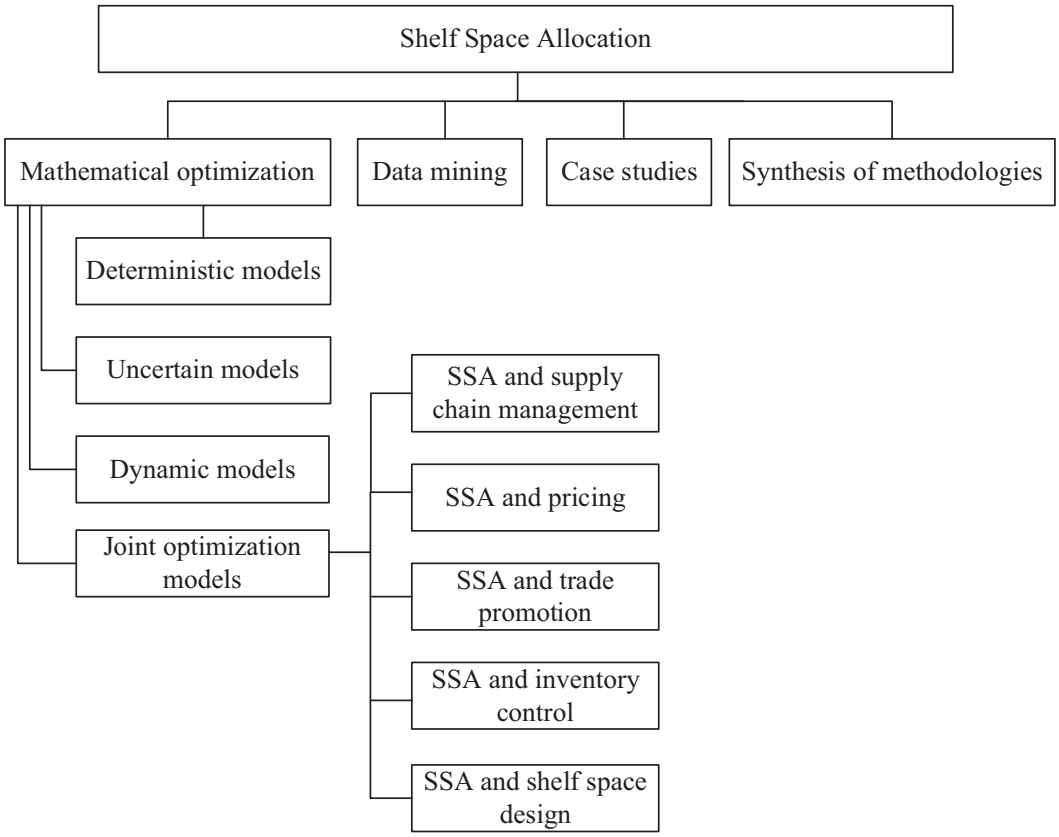

FIGURE 3. Shelf space allocation literature classification.

- **Uncertain models:** real-world retail environments often involve unpredictable variables such as fluctuating consumer demand. By integrating uncertainty into optimization models, retailers can develop more resilient strategies that accommodate variability, ultimately reducing stockouts and overstock situations.
- **Dynamic models:** this subcategory emphasizes the importance of time in decision-making. Retail environments are not static; thus, dynamic models adapt to changing conditions and allow retailers to respond proactively to seasonality and market trends, enhancing long-term effectiveness.
- **Joint optimization models:** the retail landscape is interconnected, where decisions in one area often impact another. This category focuses on the integration of different functions to achieve overall optimization:
  - *SSA and supply chain management*: understanding shelf space allocation within the broader context of supply chains is crucial. This subcategory investigates how upstream and downstream activities influence shelf space strategies, promoting a holistic view that can lead to more efficient resource utilization across the supply chain.
  - *SSA and pricing*: this subcategory examines how pricing strategies can be aligned with shelf space decisions. An optimized pricing model that considers shelf placement can enhance sales and profitability, highlighting the synergy between these two critical areas.
  - *SSA and trade promotion*: trade promotions are a significant part of retail strategy. By understanding how shelf space allocation affects and is affected by these promotions, retailers can design more effective marketing campaigns that drive sales while improving space efficiency.

- *SSA and inventory control*: efficient inventory management is key to maintaining optimal shelf space. This subcategory underscores the interdependence between inventory levels and shelf allocation, ensuring that products are available where they are needed, thus maximizing sales potential.
- *SSA and shelf space design*: this area focuses on the physical arrangement of products. Understanding consumer behavior in relation to shelf layout is crucial for enhancing visibility and accessibility, leading to improved shopper experience and increased sales.

Data mining

As retailers collect vast amounts of data, the ability to analyze this information becomes paramount Ziari and Sajadieh [117]. The data mining subcategory emphasizes:

Implementing large datasets to derive insights about consumer preferences, purchasing patterns, and market trends.

Implementing data-driven decision-making processes that enhance the precision of shelf space allocation Ziari [116]. This category encourages a shift from intuition-based decisions to informed strategies that reflect actual consumer behavior.

Case studies

This category serves a dual purpose: It illustrates real-world applications of theoretical models, providing valuable insights into the practical challenges retailers face when implementing shelf space strategies. Analyzing successes and pitfalls in diverse contexts offers lessons that can enhance understanding and improve future SSA approaches, making it a crucial reference point for both researchers and practitioners.

Synthesis of methodologies

The final category encompasses comprehensive evaluations of various SSA strategies:

It includes empirical studies that assess effectiveness and identify best practices, enabling retailers to benchmark their performance against industry standards.

Robust analytical frameworks within this category help synthesize findings across studies, providing a clearer picture of effective SSA practices, facilitating research progress, and guiding future investigations.

In conclusion, the classification into these five categories with detailed subcategories is designed to systematically address the multifaceted nature of SSA literature. This structure enhances clarity, encourages further research, and fosters a deeper understanding of the complexities involved in shelf space allocation, ultimately contributing to both academic inquiry and practical applications in retail settings.

The division of shelf space allocation (SSA) literature into five distinct categories is grounded in the diverse and complex nature of retail environments. By segmenting the literature into these categories, we can more effectively map the multifaceted dynamics at play in retail, ensuring that we account for the variety of factors influencing shelf space decisions, from mathematical principles to practical applications. The classification recognizes that SSA is not a one-size-fits-all discipline. The variety of approaches – ranging from mathematical optimization to case studies – reflects the need for strategies that consider the unique characteristics and constraints of different retail contexts. The integration of simultaneous optimization underscores the necessity of aligning various retail functions. This interconnected approach acknowledges that effective shelf space allocation cannot be isolated from aspects like pricing, promotions, and inventory control, thereby fostering a comprehensive view of retail strategy. Moreover, including case studies and empirical analyses within the classification serves to bridge theoretical concepts with real-world applications. This connection enhances the practical relevance of SSA literature, providing actionable insights that practitioners can implement. And eventually, this structured approach not only clarifies existing knowledge but also highlights gaps in the literature, encouraging further investigation and innovation in SSA methods. By delineating these categories, researchers can easily identify areas that require more exploration and thus contribute to the advancement of the field. In summary, the rationale for this categorization lies in its ability to systematically address the complexities of shelf space allocation, facilitating a deeper understanding and more effective application of strategies within the retail landscape.

3.1. Mathematical optimization

Here, we describe a basic shelf-space allocation problem. Different criteria are available as the objectives of SSA. The most frequently used objectives are maximizing total sales/profit, minimizing total costs, and maximizing the productivity of the retailer. One of the general models in the literature is the one introduced by Yang [110]. The objective function is maximizing the total profit of the stores' entire products subject to store capacity constraints. The following nomenclature is used for developing this basic model.

Sets	
i	Set of products
k	Set of shelves
Parameters	
T_k	Length of each shelf
p_{ik}	Profit of product i on shelf k
U_i	Upper bound for the number of faces for product i
L_i	Lower bound for the number of faces for product i
a_i	The facing of product i
Decision variables	
x_{ik}	The number of facings of product i on shelf k

The model now can be written as follows:

$$\text{Max } P = \sum_{i=1}^{n}\sum_{k=1}^{m} p_{ik}x_{ik}$$

s.t.

$$\sum_{i=1}^{n} a_i x_{ik} \leq T_k \qquad \forall k = 1, \cdots, m$$

$$L_i \leq \sum_{k=1}^{m} x_{ik} \leq U_i \qquad \forall i = 1, \cdots, n$$

$$x_{ik} \in N \cup \{0\} \qquad \forall i = 1, \cdots, n, \quad \forall k = 1, \cdots, m.$$

Total profit (P) is obtained by multiplying the number of facings of each product (x_{ik}) by the profit of that product [19]. The process is repeated to find the total retailer profit of all n products on all m shelves. The objective of the problem is to find the best combinations of products and the shelf space allocated. Based on the first constraint, the summation of the facing of each product (a_i) should not exceed the length of each shelf (T_k). Moreover, an upper bound and lower bound for the number of faces of each product are defined (*i.e.*, U_i and L_i, respectively). As can be seen, the problem is formulated as an integer programming model because the retailer is interested in obtaining integer numbers of product facings. In order to find how this optimization must work, pay attention to the Figure 4. It can be seen that the problem is finding the most appropriate number of products based on its characteristics including the facing, size, preference, bran and etc. Moreover, the number of facing and how the inventory of a shelf can be calculated in illustrated in Figure 5.

Papers in SSA literature can be classified into different categories based on decision variables, structure of retailing supply chain, etc.

3.1.1. Deterministic models

Most papers in the retail shelf space allocation literature focus on mathematical models. Anderson and Amato [5] is one of the first mathematical models in the literature. They simultaneously determine the assortment composition and the allocation of the available display area. Then, Hansen and Heinsbroek [52] propose a

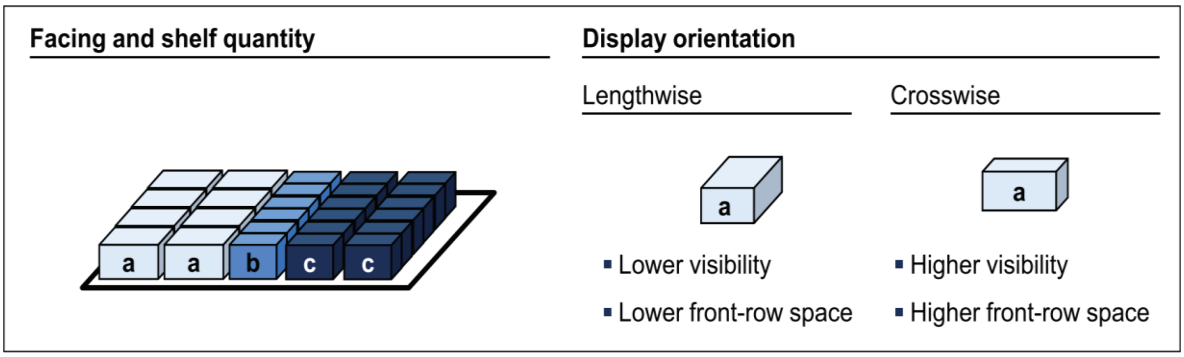

FIGURE 4. Representation of shelf facing, quantity and orientation of display [59].

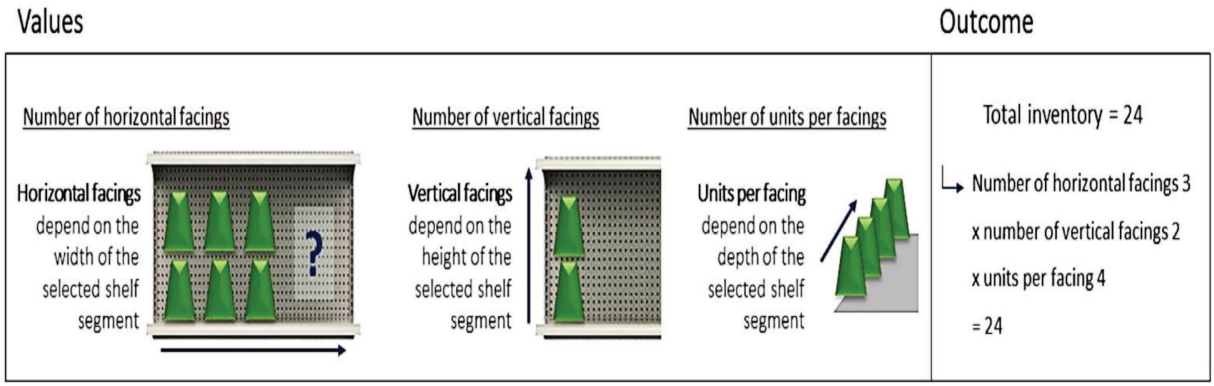

FIGURE 5. Representation of different options for number of facings [62].

model of simultaneously optimizing the selection among a given set of items to be sold in a supermarket and the allocation of available shelf space to them. To address shelf space elasticities, Corstjens and Doyle [26] develop a general mathematical model where both elasticities as well as cross-space elasticities are considered. The model is solved using signomial geometrical programming. They show that the results of their model are quite different from previous simpler models. Then, their work is extended by Bultez and Naert [16] to another model in which the demand interdependencies prevailing across and within product group are incorporated. Bultez *et al.* [17] more generalize the model Bultez and Naert [16] to integrate the diversity of substitution effects that can be inflected by package-size purchasing habits, brand loyalty, or preference for a precise variety. Then, Bookbinder and Zarour [12] employ the model of Corstjens and Doyle [26] and add it by integrating the concept of Direct Product Profitability (DPP) within a shelf-space allocation model. They show how DPP may be employed within the SSA optimization models.

Borin *et al.* [14] formulate a problem to optimize assortment and allocation of space to the products. They use a Simulated Annealing (SA) search algorithm to find a good shelf space allocation. Borin and Farris [13] later analyze the error degree that may be used in the estimates of parameters. Yang [110] develops a heuristic algorithm based on the linkage between the multi-constraint Knapsack problem and the SSA problem. The ranking order of the profit per displayed length of each item is used as a priority index in the shelf space allocation process.

To complete previous gaps, Lim *et al.* [78] examine a linear model for optimizing the SSA problem. They extend that model to address other requirements such as nonlinear profit function and product grouping. They

also explore the potential of certain metaheuristics to solve basic SSA problems introduced by Corstjens and Doyle [26, 27] and Yang and Chen [111]. Hwang *et al.* [64] extends the SSA literature to the case in which the retailer sells several brands of products through multi-level shelves, where the shelf level has a significant effect on sales. Reyes and Frazier [96] develop a nonlinear integer goal programming SSA model that takes into consideration two conflicting objectives, *i.e.*, customer service factors and profitability.

In addition, Gajjar and Adil [42] contribute to the literature by reformulating the existing nonlinear shelf space allocation model to an integer programming model by employing piecewise linearization. They also develop a heuristic algorithm to produce near-optimal solutions for randomly generated instances. Hansen *et al.* [53] integrate product cross-elasticity, horizontal and vertical location effects, and also nonlinear profit function into the basic shelf-space allocation models. They introduce a linear programming formulation of the nonlinear profit function to optimize SSA decisions. Gajjar and Adil [44] propose three heuristics to solve the linear shelf-space allocation model given by Yang and Chen [111]. Irion *et al.* [66] extend the work of Corstjens and Doyle [26] in three directions, *i.e.*, allocating shelf space to items as several facing; simultaneously optimizing assortment and shelf space decisions; and modeling cost elements including replenishment and assortment costs. Bai *et al.* [8] present a two-dimensional shelf space allocation model in which both shelf length and height are incorporated in allocation decisions.

Castelli and Vanneschi [20] develop a hybrid Genetic Algorithm with Variable Neighborhood Search (GA-VNS) algorithm for the SSA problem. Recently, Leng *et al.* [76] generalize the model of Corstjens and Doyle [26] to estimate the demand for each item category in a problem known as space exchange. They obtain decisions regarding shelf-space allocation in Nash equilibrium for the case in which several retailers can exchange their excess shelf spaces to improve their space resource utilization. Geismar *et al.* [48] introduce a revenue maximization model in retail stores where a vertical as well as a horizontal component are considered for the shape of each product's presentation. Zhou and Piramuthu [115] develop a shelf-space allocation model when ticket-switching behavior is present at a store that stocks cheap as well as expensive items. Flamand *et al.* [40] integrate shelf and aisle allocation, shelf-space allocation, and location of products within shelves in a model is jointly optimized. Products are pre-clustered and if products have high affinity/relations, they should be allocated to the same aisle/shelf while there are some products that should not be allocated to the same shelf because of low affinity. Traffic within aisles is addressed which affects product visibility and as a result, increases in impulse buying chances. Later on, Bianchi-Aguiar *et al.* [11] develop a ga to be used by the space management team to find the optimal lay out of products on each shelf. The proposed approach is capable of creating a trade-off between optimization and customization by composing the analytical strength and including different kinds of merchandising rules.

After that, Moussaoui *et al.* [86] presents a systematic review of drivers of On Shelf Availability (OSA), identifying the future research directions, and finding suitable alternatives for OSA. Then, they propose five streams of drivers in OSA. Furthermore, Flamand *et al.* [41] develop a mixed-integer programming model to show that SSA's expected profit depends on both the shelf attractiveness and the profitability of products. They embedded their paper by an optimization-based heuristic method. The method works on initial feasible solutions which are refined by re-optimized shelf subsets. Finally, they implemented the proposed model in a grocery store to show the method application and its accuracy.

Almost all the mentioned papers have assumed that there is a closed-form function for demand while there are some situations in which this assumption is not applicable. For example, when the demand of products depends upon the time customers spend in a store, whether a specific aisle or shelf is visited by customers or not, whether the store is crowded and whether many customers are visiting the store simultaneously or not, in this situation, it is not possible to develop a closed-form function to find demand of products and simulation can be an effective tool to find demand. Therefore, a simulation-based optimization approach can be applied to this type of problem. Moreover, to make the problem more realistic and applicable in real-world problems, it would be useful to develop two or three – dimensional shelf space allocation models which is studied by Murray *et al.* [87], Bai *et al.* [8], Geismar *et al.* [48], Rabbani *et al.* [93], and Zheng *et al.* [114].

3.1.2. Uncertain models

The research by Amit *et al.* [2] is one of the rare studies on the optimization of the number of products stocked on retailer shelves under stochastic conditions. They obtain the shelf-space stocking policy under stochastic demand so that the retailer profit is maximized. It is assumed that the demand is influenced by both endogenous and exogenous factors.

As it is obvious, only Amit *et al.* [2] have studied SSA problems under stochastic conditions (in addition Lotfi and Torabi [79] consider the parameters of the problem ambiguous and use fuzzy mathematical programming) while in real-world problem, it is unlikely to have exact value of parameters. Stochastic programming which uses historical data and Robust Optimization (RO) that uses the interval of the value of parameters (or scenarios for maximum regret approach) is potentially useful to model the SSA problems. It might be beneficial to integrate shelf/aisle allocation and shelf-space allocation problems by using two-stage stochastic programming. Furthermore, Hübner and Schaal [60] develop the first model which considers stochastic demand in shelf-space planning problems especially for promotional or temporary products. They consider demand uncertainty for their model like what was considered by Alfares [1] in relation to stochastic models. In addition, they clearly show that demand is derived by impulses, the more is depended on spaces and also is subject to variations. Their proposed model and solution approach result in higher profit when shelf space planning and stochastic demands interact. In the same way, Hübner and Schaal [59] study an SSA problem by considering stochastic demand that is also space-elastic. Then, they solve the proposed model by developing a novel heuristic method to obtain near-optimal solutions effectively. They applied the model for both perishable and non-perishable products and proved that this approach can help retailers to boost their profit and market situation.

3.1.3. Dynamic models

Most of the existing models in the literature focus on static optimization. For the first time, Corstjens and Doyle [27] extend the static model to incorporate market dynamics such as new tastes and changing product life cycles. Zufryden [118] then develops a dynamic programming model considering general objective-function specifications, *i.e.*, space elasticity, demand-related marketing variables, and the costs of sales.

Gajjar and Adil [43] also, develop dynamic programming heuristic to find a near optimal solution due to the NP-hardness of the non-linear model of SSA. Gajjar and Adil [45] extend Gajjar and Adil [43] with inventory replenishment. They considered a continuous-review model for controlling inventory levels. A dynamic programming algorithm and local search heuristic is used to solve the proposed model. Gilland and Heese [50] optimize shelf space allocation under dynamic substitution assumption for two products in a retail store in which customers sequentially arrive and serve themselves from the products on the shelf.

It is worth mentioning that some papers in this category are not dynamic but they have used dynamic programming. Although markets in recent years have been highly dynamic, just three papers have considered the dynamic structure of the model. Coping with the stochastic nature of data as well as the dynamicity of markets, a new direction for the development of stochastic dynamic programming and reinforcement learning approaches can be considered. Also, their approaches can provide online optimization of decisions in retail management.

3.1.4. Joint optimization models

Several papers in the literature study the situations in which the shelf-space allocation decision is made in alliance with other operational decisions such as pricing, trade promotion, inventory policies, shelf-space design, etc. The reason is that individually managing the shelf-space allocation, product assortment, product display area selection, inventory control and other retailing operations will result in sub-optimal solutions overall retailer's profit. Thus, some researchers have developed models to integrate these operations to increase the retailer total profitability.

3.1.4.1. SSA and supply chain management. In recent years, several publications have addressed the SSA problem by considering multiple firms in a supply chain rather than a single retailer. Martín-Herrán *et al.* [84] is one of them that studies a supply chain consisting of two competing manufacturers and a retailer. The

manufacturers optimize advertising decisions, while the retailer decides on allocating the shelf space to both brands. The problem is formulated as a Stackelberg differential game which is played over an infinite horizon, assuming two manufacturers and one retailer as leaders and followers, respectively. Amrouche and Zaccour [4] study a supply chain with one manufacturer and a single customer who owns private-label products. Two models are developed such that in the first model, the manufacturer does not offer any incentive and Nash equilibrium is the outcome of the game while in the second model, the manufacturer offers wholesale price based on shelf space allocated to its products at the retailer's shelves. The manufacturer (retailer) is considered the leader (follower) and Stackelberg game is used to model the game. Numerical results reveal it is profitable that the manufacturer offers incentives.

Hariga and Al-Ahmari [54] develop profit-maximization models to study a supply chain operating under consignment stock [33] and Vendor-Managed Inventory (VMI) agreement, where the manufacturer decides about the reorder point, the size of each order, and the quantity to be displayed on the shelves. Wang and Gerchak [108] study a model in which there is a single manufacturer and a single retailer. Demand at retailers depends on inventory at shelf space and is increasing and concave in terms of inventory. To coordinate the supply chain, the manufacturer adjusts wholesale pricing and inventory subsidy. They develop this model to a single manufacturer and two retailers and developed two models. In the first model, total demand is determined based on a total inventory of both retailers and demand at each retailer is defined based on relative inventory. In the second model, customers select retail based on their relative inventory and then purchase according to individual retailer's inventory. More recently, Kuiti *et al.* [73] studied a shelf space, product design and transportation decisions for a green supply chain problem. They presented a supply chain including a manufacturer and a retailer where the manufacturer tries to green the supply chain using pack-size and transportation cost reduction.

Most papers have considered a supply chain with one or two manufacturers and retailers. However, in real-world retailing, the number of manufacturers is much more than two and it would be more realistic to develop a model in this situation. Also, it would be more realistic to consider a supply chain with multiple manufacturers and chain stores in which different branches of chain store have different demands, sales, and assortments. In this situation, manufacturers should find incentives for each branch of a chain store. Moreover, it makes the problem more beneficial to study competition between different chain stores and their cooperation with the same manufacturers.

3.1.4.2. SSA and pricing. Martín-Herrán *et al.* [84] seems to be one of the first researches that incorporates pricing decisions into SSA. They consider that prices and the share of shelf space allocated to products are decisions to be made by the supply chain players including two competing manufacturers and one retailer. They arrive at the conclusion that the wholesale prices of products are considerably related to their shelf-space share. Later, Amrouche and Zaccour [3] simultaneously tackled pricing strategies and SSA. They develop a game-theoretic model to optimize the manufacturer's price as well as the amount of retailer's shelf space allocated to the products and their selling prices. More recently, Nakade and Ikeuchi [89] work on a pricing model which first obtains the optimal price and ordering amount of products, then decides on the appropriate location of items in a shelf space based on their clearance. Murray *et al.* [87] propose a model that jointly optimizes display orientations, display-facing areas, shelf-space locations, and product prices in a product category, considering both the width and height of each shelf. Nafari and Shahrabi [88] also develop an approach to optimally select and price the products and allocate them to shelf space with consideration of their prices. They used association rules as data mining techniques to find relationships between products regarding their prices. Martínez-de-Albéniz and Roels [85] build a SSA model on the work of Corstjens and Doyle [26] by characterizing competitive dynamics caused by the lack of retail shelf space in a multi-supplier single-retail supply chain, when the selling prices are both endogenous and exogenous.

Li *et al.* [77] introduce game theoretic models to simultaneously optimize pricing as well as shelf-space allocation decisions under competition between two retailers. Policies are optimized for substitutable products in two manufacturers- two retailers supply chain for the cases in which supply and demand are balanced or unbalanced. Leng *et al.* [75] use the hoteling model as well as a non-cooperative game to optimize the selling price and allocated space under the space-exchange strategy. In this strategy two retailers can exchange their

shelf space to improve their access to more consumers in more locations instead of opening new retail stores. Kuo and Yang [74] also develop a game-theoretical SSA model for a supply chain with a single manufacturer and single retailer. The retailer can sell the shelf space to a competing national brand or launch its store brand. It is assumed that first, the retailer decides on the category shelf space. Then, the manufacturer determines the shelf-space fee to buy the space and its wholesale price for placing its national brand products. Finally, the retailer sets the selling price of each brand. Recently, Wang *et al.* [109] introduce a model to simultaneously optimize the shelf-space size as well as the retail and wholesale pricing decisions in a supply chain with two manufacturers and one retailer. They assumed that the items have unit production costs, non-symmetric market potentials, and cross-price sensitivity parameters. Zhao *et al.* [113] proposed a joint shelf space and pricing allocation optimization model in a retailer-dominated supply chain considering consignment-based revenue-sharing contracts. The developed model includes a two-echelon supply chain (two manufacturers and a dominant retailer) delivering substitutable products in a market with shelf space and price-dependent demand. Taleizadeh *et al.* [102] presented a new model to determine the size, warranty and price of products in a two-echelon supply chain. This research involves adopting mixed strategies. Reza [97] investigated a revenue sharing contract. As demand affects the inventory systems, they tried to incorporate these impacts in a shelf-space allocation problem defining a new demand function. Centralized, decentralized and coordinated structures are examined considering competition to derive the best solutions.

Almost all papers have assumed a constant quality for products over time while this assumption is not realistic for some products such as grocery, diary etc. In this type of product, the quality of products degrades over time and is would be interesting to optimize pricing of fresh products as well as products with quality degradation to maximize total profit. Also, pricing decisions for products with seasonal demand or the ones whose sales increase/decrease on specific days like holidays would be interesting.

3.1.4.3. SSA and trade promotion. Tsao *et al.* [104] introduce a category SSA framework with trade allowance where a Stackelberg game is developed to present the interaction between multiple manufacturers and one retailer. The model optimizes the retailer decisions (assortment decision, promotion level, and SSA) and manufacturers' decisions (trade allowances).

3.1.4.4. SSA and inventory control. Urban [105] is one of the first researches that responses to the need for integrative research in operations and marketing management. He integrates two different streams of research that are almost interdependent in retail organizations, *i.e.*, SSA and inventory management. The model introduced optimizes three decision variables: reorder point, order quantity, and display quantity. Maiti and Maiti [81] develop a multi-product model in which products are either substitute or complementary. Therefore, the demands of the products depend on the unit price of the respective items, the amount of stock on the shelf space, and the presence of substitute or complementary items. Decision variables are reorder point, order quantity, displayed inventory level, and selling price which are optimized to maximize total profit.

Hariga *et al.* [55] also introduce an integrated model to jointly optimize inventory replenishment decisions as well as the product assortment, and shelf space allocations so that the retailer's profit is maximized. Ramaseshan *et al.* [94] propose a decision support tool for category management to generate optimal SSA, product assortment, review period, and order quantity to maximize the total net profit of an independent franchise retailer. Bai and Kendall [6] for the first time propose a single-period inventory and SSA model for fresh items with a very short lifetime, *i.e.*, the demand rate depends on both the inventory displayed on the shelf and the items' freshness condition. The decision variables are order quantity, shelf space, and the surplus quantity at the end of the planning cycle. Bai *et al.* [7] developed and compared several heuristics and meta-heuristics as alternatives for the generalized reduced gradient algorithm introduced by Bai and Kendall [6]. Later on, Hübner and Kuhn [56] aim to answer a retailer's fundamental question namely, how much product should be ordered and how much shelf space does it need. They mostly implemented Operation Research (OR) concepts to examine cross-space impacts and replenishment limitations on profit and reveal improving profit levels by customer behavior.

In a more realistic model, Piramuthu and Zhou [92] develop Bai and Kendall's model and studied inventory management of perishable products in which the demand depends on shelf space allocated as well as the quality of each product. To determine the quality of products, Radio Frequency Identification (RFID) tags are used

which transfer data like temperature. The lower quality, the lower demand and if the quality reaches a threshold, there would not be any demand for that product. Genetic Algorithm (GA) is used to solve the model.

Gao and Meng [46] introduce a problem of jointly optimizing product variety, inventory control, and shelf-space allocation to maximize total profits, where the rate of customer demand is formulated as a function of time, promotion and shelf space. Baron *et al.* [9] later study the joint optimization of inventory (*i.e.*, reorder points and order quantities) and SSA decisions for multiple products with a demand rate that depends on both above factors to maximize the expected profit rate of the retailer. As discussed earlier, Hariga and Al-Ahmari [54] also develop models where the supplier decides about the quantity to be displayed on the shelves size as well as inventory control decisions, *i.e.*, order size and the reorder point.

Lotfi *et al.* [80] develop a model considering replenishment planning in which all items with the same brand can be jointly replenished, inventory control, and shelf-space allocated to each product. Two conflicting objective functions profitability and customer service are optimized through a weighted goal programming approach. In this model, the demand of each product depends on the number of allocated faces as well as changes in itself and its substitution prices.

A similar study is carried out by Lotfi and Torabi [79] and they consider shelf-space allocation, inventory and replenishment planning of a set of products with different brands. It is assumed that there are loyal and disloyal customers and the demand of loyal customers is constant and should always be satisfied while that of disloyal customers can be changed based on shelf faces allocated to each product and brand as well as its shelf level. Three objective functions comprise total profit, service level, and space utilization; and because of the imprecise nature of data, fuzzy goal programming is used to aggregate these objective functions.

Recently, the impact of customers' shopping paths on shelf-space planning has been studied by Ghazavi and Lotfi [49]. The total demand for products consists of fixed loyal customers' demand and variable which is defined for potential customers. The store is divided into different zones with different conditions and variable demand depends on some factors such as zone and level of shelves, product substitution impact. They also consider optimization of inventory, order quantity and number of replenishments during the planning horizon, as well as rearranging the shelf spaces in different periods. Simulation-based optimization Genetic Algorithm Independent Component Analysis (GA-ICA) is used to solve the proposed model. Zhao *et al.* [112] studied a model in which demand is a function of shelf-space allocation, display location, and spatial relationships among items. A spatial relationship defines the increase in the sales of complementary items or decreases in that of substitution items based on adjacency. In this paper, display location, shelf space allocation, and replenishment cycles are determined to maximize total profit. Two models are developed such that in the first model all items are individually replenished while in the second model, items can be jointly replenished. A multi-stage SA-based hyper heuristic algorithm is developed to solve the model. Later on, the joint optimization of SSA and replenishment policies is the other critical and effective issue which is performed [59]. They tried to find the optimal facings number, product orientation and number of orders considering space-elasticity. Then, the model was applied to a grocery retailer which resulted in a 29% higher profit. Moreover, applying the model to random data shows that the model can be solved in large scales with short execution times.

Moreover, as inventory management of fresh products is different from other products because of quality degradation, it is so important to develop some specific SSA models while just two papers have addressed this issue [6, 92]. Düsterhöft *et al.* [36] worked on shelf space allocation problems while optimizing replenishment policy. They proposed a first shelf-space model capable of modeling three-dimensional shelf-space features. And finally, they applied the model for a German grocery retailer to show the model application for profit maximization. Kim and Moon [70] presented a shelf space, product selection and replenishment policy for a retailer. The developed model is based on a two-dimensional display space considering space dimensions, product category and the other factors influencing customer demand. The model finally is solved using two heuristic methods. Sajadi and Ahmadi [99] investigated an integrated optimization of inventory decisions and shelf-space allocation problems. They aimed to deliver the best set of selected suppliers for perishable products in a real-world industrial case problem. Hübner and Kuhn [58] tried to develop a new framework capable of optimizing shelf space allocation decisions and replenishment policies. To this aim, they developed an integrated novel

model particularly responding to shelf-space allocation assignments and an assortment of fast-moving products. They applied the model to a real-world case and proved the improvements in profit.

3.1.4.5. SSA and shelf space design. Hwang *et al.* [64] develop an integrated model to simultaneously optimize the shelf space design and product allocation considering space elasticity and location effect on demand with the objective function of maximizing the retailer's total profit. They assume that the demand rate is a function of product amount and the location displayed within shelf space. Russell and Urban [98] introduce a discrete as well as a continuous shelf-management model in which the product's location is considered as a decision variable. Moreover, the products of a family are located together, *i.e.*, maintaining product family integrity as a unit is taken into consideration.

Beed *et al.* [10] studied a shelf space allocation problem and developed a hyper-heuristic approach for allocating the shelf space to products while maximizing the retailer's profit. They proved that the developed hyper-heuristic algorithm improves the shelf space allocation problem and enhances the retailer's profit. Vincent *et al.* [107] presented a new model to determine the best combination of products in each space for maximizing total profit. The main issue in their paper was allocating the right product in the right space with the right size.

As the literature lacks simultaneous optimization of shelf dimensioning and product allocation, Hübner *et al.* [62] proposed a new approach to cover this gap. They proposed an integrated approach to quantify the facing of each product, shelf space, size and number of each space. Similarly, Ostermeier *et al.* [91] presented a novel store-wide shelf space problem capable of optimizing shelf space assignment for each category considering retail profits. Gecili and Parikh [47] tried to analyze the mismatch of product shape and the dimensions if the rack in a joint shelf dimension and product allocation optimization problem. They solved the developed model using a decomposition-base method using particle swarm optimization. Czerniachowska *et al.* [30] studied a shelf-space allocation problem considering visible horizontal and vertical categories. They used both linear and non-linear mathematical optimization and solved it using a CPLEX solver. Later on, Czerniachowska *et al.* [32] worked on dimensioning the rack storage and shelf space for a distribution center considering zone picking. They applied two heuristics for solving the model and archived the results in the CPLEX solver again. Although their previous attempts brought noticeable advantages, Czerniachowska *et al.* [29] presented a new modified GA for the profit maximization of a shelf space allocation model. It worths mentioning that, they proved the validity of this algorithm on 25 test cases and showed its application and superiority. Chen *et al.* [23] focused on a retail industry evolving omnichannel paradigm. They tried to optimize the selection of products and shelf space allocation decisions. To solve the proposed model, they used a reformulation linearization technique and derived managerial insights.

To make shelf space design and SSA model more beneficial, it is interesting to consider shelf level, hand level and position of each shelf in the store in an integrated model. In this type of model, the demand of products depends not only on shelf level in the aisle and whether shelves are located between knees and eyes, but also on whether shelves are located close to enter and exit doors of the store or not.

3.2. Data mining

The progress of information technology makes retail stores collect transaction data from the point of sale [51] at a low cost. Employing data mining techniques this large volume of data can then be extracted to support the shelf space allocation decisions. Chen *et al.* [22] is the first one that takes advantage of data mining in SSA problems. They use data mining to identify and classify the effects of the relationship between the spatial distance of exhibited items and their sales in a retail store. Chen and Lin [21] utilize multi-level association rule mining to optimize both product assortment as well as product allocation problems in retailing. This approach based on data mining makes simultaneous decisions about the items to be stored (assortment planning), the amount of shelf space to be allocated to the selected items and also the place used to display them. Moreover, as mentioned above, Nafari and Shahrabi [88] employ data mining techniques to optimize product assortment, price selection and SSA at once. Recently, Tsai and Huang [103] proposed a solution approach to the SSA problem employing a data mining technique considering moving behaviors as well as customer purchases. Their technique comprises data collection, pre-processing, data mining and the assignment of shelf space. They classify

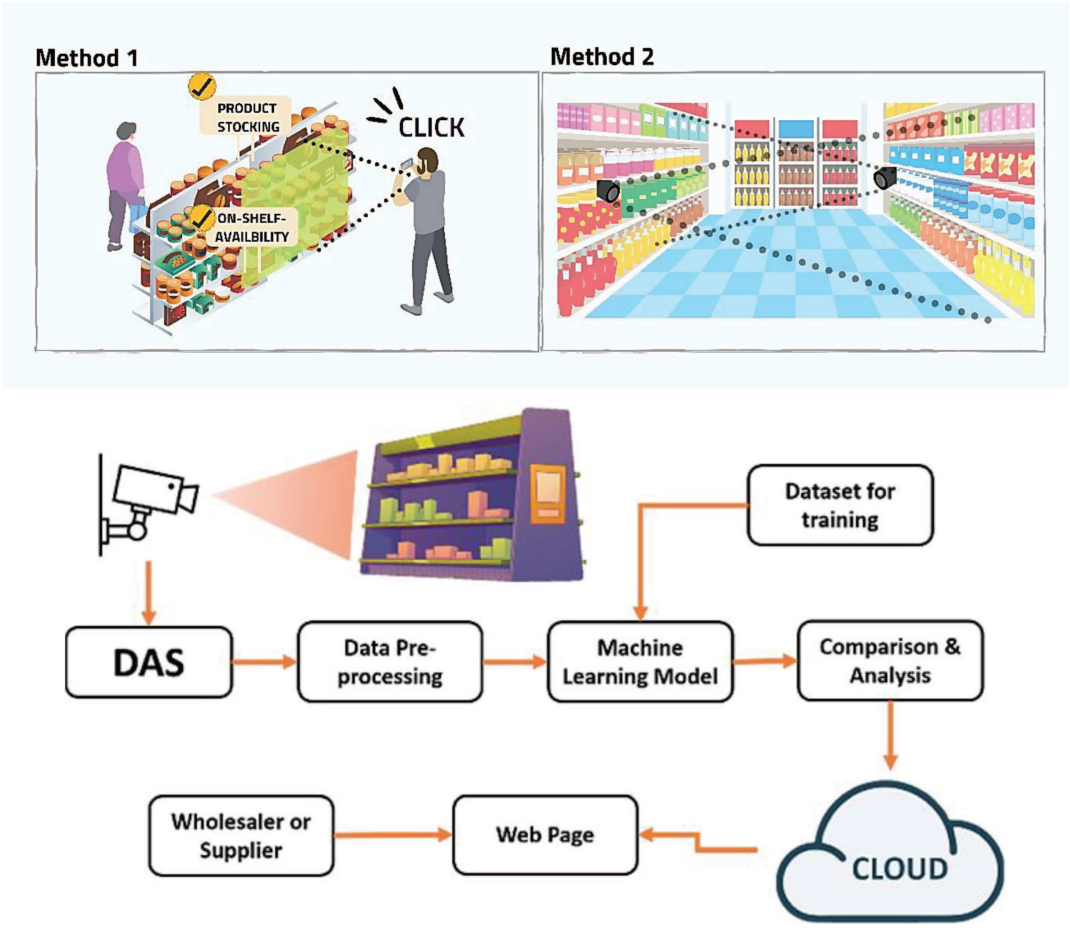

FIGURE 6. Illustration of data mining application in SSA [69].

products in the retail store as major, minor and trivial products and conclude that only items in the minor group are relocated to make sure that the consumers can follow their chosen shopping paths. Zheng *et al.* [114] developed a data-driven two-dimensional shelf-space allocation problem using a hybrid Genetic algorithm. The proposed model is mainly capable of replacing infeasible solutions with newly generated ones. They also proved the model's validity in different industrial real-world cases. Similarly, Edirisinghe and Munson [37] used data-driven rearrangements based on the profitability and affinities of products. They studied brick-and-mortar retailers competing on an online platform. The innovative concept of the proposed model was on the rule mining and assignments of departments.

With advances in IT, it is much easier to collect transactional data and based on these data and loyalty cards, customers can be clustered and classified; and customized marketing, promotion, and pricing decisions be made easier for each category of customers, and therefore, specific incentives can be offered. Moreover, as it can be seen in Figure 6, retailers can collect data through customers' smartphones connected to the Wi-Fi of the store. They can find how customers visit the store, which floors are visited by each customer, which part of the store is visited more than others, as well as the sequence of aisles and shelves visited by customers. Retailers can use this information to re-arrange their aisles and shelves to attract more customers.

3.3. Case studies

Although there is considerable academic attention to the subject of SSA, not many empirical studies have been made. We provide detailed discussions of some of these empirical papers and also introduce additional studies for further reading. The first case study in sales-to-shelf facing relationships seems to be an investigation by Kotzan and Evanson [72] that makes some experiential studies for drug store products from US chain drug stores. They find that there is a considerable relationship between sales and shelf facing. Brown and Lee [15] also, show how the shelf space can be allocated to different juice product items in some US grocery stores in a retailer profit maximizing framework. Chiang and Wilcox [24] study how shelf-space allocation and product variety decisions are related to an Indianapolis-based grocery retailer called Marsh Supermarkets. Later, Desmet and Renaudin [34] use several chain store databases at the product category level to present an empirical estimate of shelf space elasticity. They show that direct shelf-space elasticity is not zero for several product categories. They also conclude that higher space elasticity is for product categories typical of impulse buying. Nogales and Suarez [90] study the space used by the private and national labels on the shelves of Spanish retail stores. They use direct shelf observation to explain that issue through a case study. To establish any relationship with shelf space management, other aspects including product assortment, promotions and prices are also analyzed. Furthermore, Koschat [71] presents empirical evidence from US magazine publishers and shows that the demand for a specific brand decreases as its inventory on-shelf decreases. They show that among the six above-mentioned case studies, four of them were done in the US and two of them were in Europe. More recently, Janeiro [68] proposed two novel biased random-key Genetic algorithms to solve the SSA problem which was inspired by a Portuguese in China. They focus on allocating the products on shelves properly and finding the optimal location and number of facings to reach more profit and finally, they prove the application of their work both in terms of constraints and execution time. In addition, one can read other case problems and simulated studies in Dreze *et al.* [35], Hansen *et al.* [53], Fadıloğlu *et al.* [39] and Eisend [38].

3.4. Synthesis of methodologies

Jajja [67] employs a system dynamics model to address the challenge of optimizing shelf space allocation (SSA) within a retail shoe store context. His findings reveal that decisions regarding SSA are significantly influenced by store-specific demographic characteristics as well as the varying demand behaviors of individual shoe brands. This emphasizes the importance of tailoring shelf space strategies to reflect the unique attributes of the retail environment, thus enhancing overall performance. Building on the foundation laid by Jajja [67], Eisend [38] makes a notable contribution to the SSA literature by presenting experiential studies that delve into shelf space elasticities. His research demonstrates a clear trend: the mean shelf space elasticities tend to increase as one moves from basic commodities to general-use products, and further to impulse purchases. This insight provides valuable implications for retailers, suggesting that different product categories require distinct allocation strategies to maximize sales potential.

In summary, to better understand the evolving landscape of shelf space allocation literature, we classify various papers based on their specific objective functions. This classification allows for a more organized and systematic analysis of existing research, facilitating the identification of trends and gaps in the literature.

Figure 7 shows the distribution of papers in each category. It worths mentioning that it might be possible that a paper belongs to more than one category.

Moreover, Table 2 represents papers published in each decade in different categories and it shows the relative importance of each category, the trend of papers over time, and categories that have rich literature or need more research. Table 3 presents a comprehensive overview of the classification of reviewed papers according to their respective objective functions, highlighting the diverse approaches taken by researchers to address the SSA challenge. A summary of mathematical models in the literature is also provided in Table 4. This table shows the major SSA papers and structures them according to some characteristics.

TABLE 2. Classification of papers based on their categories and published time period.

	1970s	1980s	1990s	2000s	2010s	2020s	Total
Assortment Selection	Anderson and Amato [5]; Hansen and Heinsbroek [52];		Borin *et al.* [14]; Chiang and Wilcox [24];	Nogales and Suarez [90]; Hariga *et al.* [55]; Chen and Lin [21]; Ramaseshan *et al.* [94]; Gao and Meng [46];	Nafari and Shahrabi [88]; Irion *et al.* [66]; Tsao *et al.* [104];		12
Deterministic Models	Anderson and Amato [5]; Hansen and Heinsbroek [52];	Corstjens and Doyle [26]; Bultez and Naert [16]; Bultez *et al.* [17];	Borin *et al.* [14]; Borin and Farris [13]; Yang and Chen [111];	Bookbinder and Zarour [12]; Yang [110]; Lim *et al.* [78]; Hwang *et al.* [63]; Reyes and Frazier [96];	Gajjar and Adil [42]; Hansen *et al.* [53]; Gajjar and Adil [44]; Irion *et al.* [66]; Bai *et al.* [8]; Castelli and Vanneschi [20]; Leng *et al.* [76]; Geismar *et al.* [48]; Zhou and Piramuthu [115]; Flamand *et al.* [40];		23
Uncertain Models					Amit *et al.* [2]; Lotfi and Torabi [79];		2
Dynamic Models		Corstjens and Doyle [27]; Zufryden [118];			Gajjar and Adil [43]; Gilland and Heese [50]; Gajjar and Adil [45];	Kuiti *et al.* [73];	6
SSA in Supply Chain				Wang and Gerchak [108]; Martín-Herrán and Taboubi [82]; Martín-Herrán *et al.* [84]; Amrouche and Zaccour [3, 4];	Martínez-de-Albéniz and Roels [85]; Hariga and Al-Ahmari [54]; Li *et al.* [77]; Kuo and Yang [74]; Leng *et al.* [75]; Tsao *et al.* [104]; Wang *et al.* [109];		13
Pricing and SSA				Martín-Herrán *et al.* [84]; Maiti and Maiti [81]; Amrouche and Zaccour [3];	Murray *et al.* [87]; Nafari and Shahrabi [88]; Martínez-de-Albéniz and Roels [85]; Li *et al.* [77]; Leng *et al.* [75]; Kuo and Yang [74]; Wang *et al.* [109];	Zhao *et al.* [113]; Taleizadeh *et al.* [102]; Reza [97]	12
Trade Promotion in SSA				Nogales and Suarez [90];	Tsao *et al.* [104];		2
Inventory Control in SSA		Urban [105];		Wang and Gerchak [108]; Maiti and Maiti [81]; Hariga *et al.* [55]; Ramaseshan *et al.* [94]; Bai and Kendall [6]; Bai *et al.* [7]; Gao and Meng [46];	Baron *et al.* [9]; Lotfi *et al.* [80]; Lotfi and Torabi [79]; Hariga and Al-Ahmari [54]; Piramuthu and Zhou [92]; Gajjar and Adil [45]; Ghazavi and Lotfi [49]; Zhao *et al.* [112];	Düsterhöft *et al.* [36]; Kim and Moon [70]; Sajadi and Ahmadi [99]; Hübner and Kuhn [58]	21
Shelf Space Design and SSA				Chen *et al.* [22]; Hwang *et al.* [64];	Urban [105]; Flamand *et al.* [40]; Zhao *et al.* [112];	Hübner *et al.* [62]; Ostermeier *et al.* [91]; Beed *et al.* [10]; Vincent *et al.* [107]; Gecili and Parikh [47]; Czerniachowska *et al.* [30]; Czerniachowska *et al.* [32]; Chen *et al.* [23]	13
Data Mining				Chen *et al.* [22]; Chen and Lin [21];	Nafari and Shahrabi [88]; Tsai and Huang [103];	Zheng *et al.* [114]; Edirisinghe and Munson [37]	6
Case Studies	Kotzan and Evanson [72];		Brown and Lee [15]; Chiang and Wilcox [24]; Desmet and Renaudin [34];	Nogales and Suarez [90]; Koschat [71];			6
Synthesis of methodologies					Jajja [67]; Eisend [38];		2

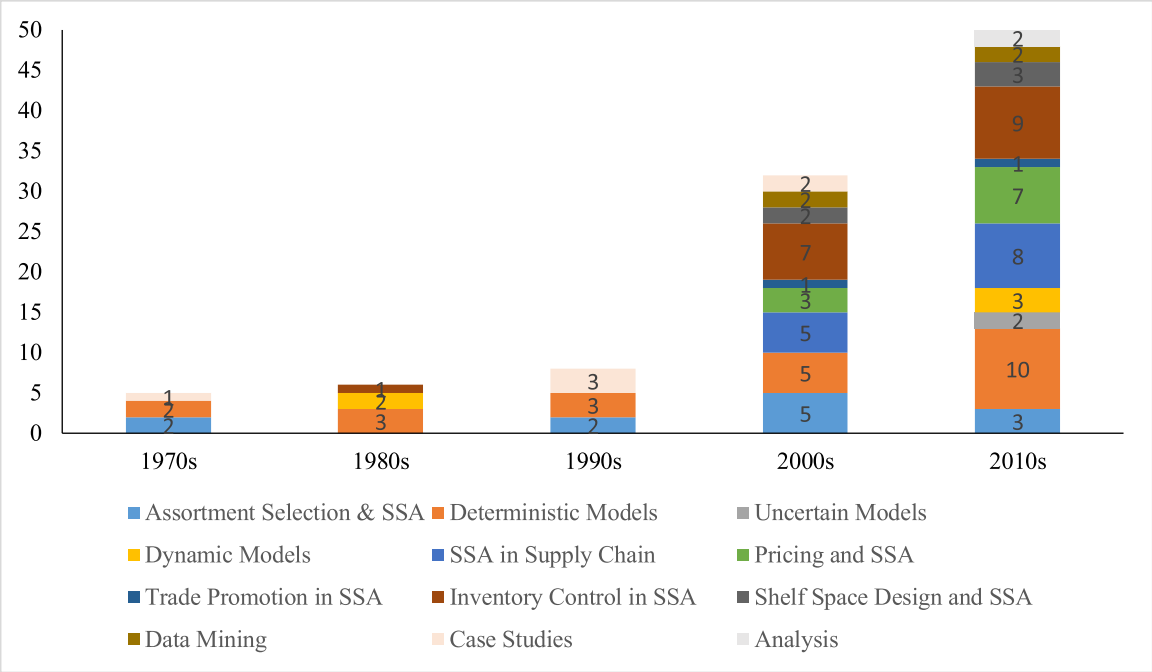

FIGURE 7. Distribution of papers in each category.

4. Conclusion and future research direction

This paper presents a review of shelf space allocation literature. Tens of papers published in the literature studied, some of them in the past decades. Overall, the SSA field is increasingly growing. However, for those looking to do research in this area, considerable works still exist for further development. While significant advancements have been made, our study identifies several main limitations and implications for future research and practical application. A predominant limitation within the SSA literature is the tendency to focus on mathematical models that optimize shelf space for single retailers. This narrow scope fails to capture the complexities of multi-channel environments where various stakeholders, including manufacturers, retailers, and distributors, must collaborate. Future research should seek to develop models that consider competitive dynamics and integrated supply chain strategies to maximize overall system profitability. Our findings indicate that empirical studies are primarily limited to specific regions, notably the US, Spain, and France. This geographic limitation can restrict the generalizability of the results. To address this gap, more empirical research across diverse cultural contexts is essential, as consumer behavior and responses to shelf space may vary significantly across different regions. Many mathematical models currently in use operate under deterministic assumptions regarding shelf space allocation. However, real-world conditions are characterized by uncertainty in parameters such as customer demand, lead times, and competitive actions. Incorporating stochastic elements into these models would enhance their realism and applicability, making them more relevant for practitioners. Another limitation is the lack of exploration into various inventory models within the SSA context. Many existing models do not account for different product characteristics or pricing strategies that could significantly influence shelf space allocation decisions. Future research should investigate these aspects to broaden the applicability of SSA models. By addressing the limitations associated with single retailer optimization, researchers can provide valuable insights into collaborative strategies among supply chain members. This approach could lead to the development of integrated decision-making frameworks that optimize shelf space at a system-wide level,

TABLE 3. Classification of papers according to objective functions.

Paper(s)	Objective function
Wang and Gerchak [108], Martín-Herrán and Taboubi [82]; Martín-Herrán *et al.* [84]; Amrouche and Zaccour [3]; Martínez-de-Albéniz and Roels [85]; Li *et al.* [77]; Kuo and Yang [74]; Hariga and Al-Ahmari [54]; Tsao *et al.* [104]; Wang *et al.* [109]; Kuiti *et al.* [73]; Zhao *et al.* [113]; Sajadi and Ahmadi [99]; Edirisinghe and Munson [37]; Taleizadeh *et al.* [102]; Chen *et al.* [23]	Maximizing manufacturer(s) and retailer(s) profits
Anderson and Amato [5]; Hansen and Heinsbroek [52]; Corstjens and Doyle [26]; Corstjens and Doyle [27]; Zufryden [118, 119]; Bultez and Naert [16]; Bultez *et al.* [17]; Borin *et al.* [14]; Urban [105]; Yang and Chen [111]; Bookbinder and Zarour [12]; Yang [110]; Lim *et al.* [78]; Hwang *et al.* [63]; Maiti and Maiti [81]; Hariga *et al.* [55]; Reyes and Frazier [96]; Ramaseshan *et al.* [94]; Bai and Kendall [6]; Bai *et al.* [7]; Gao and Meng [46]; Hwang *et al.* [64]; Raut *et al.* [95]; Murray *et al.* [87]; Russell and Urban [98]; Hansen *et al.* [53]; Gajjar and Adil [42–44]; Baron *et al.* [9]; Lotfi *et al.* [80]; Lotfi and Torabi [79]; Irion *et al.* [66]; Leng *et al.* [75]; Piramuthu and Zhou [92]; Gilland and Heese [50]; Castelli and Vanneschi [20]; Leng *et al.* [76]; Amit *et al.* [2]; Gajjar and Adil [45]; Zhao *et al.* [112]; Ghazavi and Lotfi [49]; Hübner *et al.* [62]; Ostermeier *et al.* [91]; Beed *et al.* [10]; Vincent *et al.* [107]; Düsterhöft *et al.* [36]; Kim and Moon [70]; Czerniachowska *et al.* [30]; Zheng *et al.* [114]; Hübner and Kuhn [58]; Reza [97]	Maximizing the retailer's profit
Bai *et al.* [8];	Maximizing income for retailer
Reyes and Frazier [96]; Lotfi *et al.* [80]; Lotfi and Torabi [79];	Maximizing customer service
Flamand *et al.* [40];	Maximizing impulse buying
Lotfi and Torabi [79]; Gecili and Parikh [47]; Czerniachowska *et al.* [32]	Maximizing space utilization

benefiting all parties involved. Expanding empirical studies to include diverse geographic locations will enrich the literature by providing insights into how cultural differences affect consumer behavior. Such research can inform integrated marketing and merchandising strategies that resonate with local consumers. Moreover, the integration of stochastic parameters into SSA models will not only enhance their robustness but also provide practitioners with more reliable tools for decision-making under uncertainty. This methodological advancement can drive better inventory management and supply chain efficiency. Finally, the exploration of new solution algorithms that address complex shelf-space allocation problems will further bridge the gap between theoretical research and practical application. By developing scalable and efficient algorithms, researchers can empower practitioners to tackle real-world challenges effectively. In summary, while the field of shelf space allocation is advancing, acknowledging these limitations and implications will guide future research endeavors and improve practical applications, ultimately leading to more effective and collaborative supply chain strategies. Another critical concern is that many existing models tend to simplify customer behavior by assuming static preferences, ignoring the dynamic nature of consumer choices influenced by factors such as promotions, seasonal trends, and competitive actions. This can lead to inaccuracies in shelf-space allocation that do not reflect actual purchasing behavior. Additionally, some challenges emerge for retailers when trying to implement theoretical models in fast-paced environments, where real-time data and rapid decision-making are crucial. These challenges include the integration of technology into traditional retail practices and the need for models that can adapt to frequent changes in product assortment and consumer demand. Emerging technologies like AI and machine learning

TABLE 4. Overview of mathematical models for shelf space allocation literature.

Paper(s)	Supply chain members		Decision variables							Model/solution method
	Manufacturers	Retailers	Space allocated	Wholesale price	Retailer price	Order quantity	Reorder point	Assortment	Others	
Anderson and Amato [5]		1	✓					✓		Heuristic algorithm
Hansen and Heinsbroek [52]		1	✓					✓		Heuristic algorithm
Corstjens and Doyle [26]		1	✓							Signomial programming procedure
Corstjens and Doyle [27]		1	✓							Dynamic programming
Zufryden [119]		1	✓					✓		Dynamic programming
Bultez and Naert [16]		1	✓							Heuristic algorithm
Bultez *et al.* [17]		1	✓							Heuristic algorithm
Borin *et al.* [14]		1	✓					✓		SA
Urban [105]		1	✓			✓	✓	✓		Greedy algorithm, GRG, GA
Yang and Chen [111]		1	✓							INLP, Multistage solving procedure
Bookbinder and Zarour [12]		1	✓							Nonlinear programming, Geometric programming
Yang [110]		1	✓							Heuristic algorithm
Wang and Gerchak [108]	1	1 to n	✓	✓					Inventory subsidy	Game theory
Lim *et al.* [78]		1	✓							TS, Squeaky-Wheel optimization
Hwang *et al.* [63]		1	✓			✓				Gradient search, GA
Martín-Herrán and Taboubi [82]	2	1	✓						Advertising	Game theory
Martín-Herrán *et al.* [84]	2	1	✓	✓	✓					Game theory
Maiti and Maiti [81]		1	✓		✓	✓	✓			GA, CMGA, SA, Greedy algorithm
Hariga *et al.* [55]		1	✓			✓		✓		MINLP, Lingo
Amrouche and Zaccour [3]	1	1	✓	✓	✓					Game theory
Reyes and Frazier [96]		1	✓							Nonlinear integer goal programming

TABLE 4. continued.

Paper(s)	Supply chain members		Decision variables							Model/solution method
	Manufacturers	Retailers	Space allocated	Wholesale price	Retailer price	Order quantity	Reorder point	Assortment	Others	
Ramaseshan *et al.* [94]		1	✓			✓		✓	Review period	Excel solver
Bai and Kendall [6]		1	✓			✓			Amount of surplus	GRG
Bai *et al.* [7]		1	✓			✓			Amount of surplus	SA, Greedy algorithm, Hyper heuristics
Gao and Meng [46]		1	✓			✓	✓		Product variety	Heuristic algorithm
Hwang *et al.* [64]		1	✓						Shelf space design	GA
Raut *et al.* [95]		1	✓					✓		Greedy algorithm, GA
Amrouche and Zaccour [4]	1	1	✓	✓	✓				Private label price	Game theory
Murray *et al.* [87]		1	✓		✓				Display facing areas, display orientations	MINLP, Branch-and-bound
Russell and Urban [98]		1	✓						Items' location	Greedy algorithm
Hansen *et al.* [53]		1	✓							Meta-heuristics, Simulation
Gajjar and Adil [42]		1	✓							Piecewise linearization technique, Heuristic algorithm
Martínez-de-Albéniz and Roels [85]	n	1	✓	✓	✓					Game theory
Baron *et al.* [9]		1	✓			✓	✓			Non-linear optimization package
Gajjar and Adil [44]		1	✓							Dynamic programming heuristic
Gajjar and Adil [43]		1	✓							Heuristic algorithms
Lotfi *et al.* [80]		1	✓			✓			Joint replenishment time	Weighted goal programming
Lotfi and Torabi [79]		1	✓			✓			Joint replenishment time	
Irion *et al.* [65]		1	✓					✓		Hierarchical decomposition
Irion *et al.* [66]		1	✓					✓		MINLP, Piecewise linearization technique

TABLE 4. continued.

Paper(s)	Supply chain members		Decision variables							Model/solution method
	Manufacturers	Retailers	Space allocated	Wholesale price	Retailer price	Order quantity	Reorder point	Assortment	Others	
Bai *et al.* [8]		1	✓							INLP, Heuristic approach
Li *et al.* [77]	2	2	✓	✓	✓					Game theory
Kuo and Yang [74]	1	1	✓	✓	✓					Game theory
Leng *et al.* [75]		2	✓		✓					Game theory
Hariga and Al-Ahmari [54]	1	1	✓			✓	✓			GRG
Piramuthu and Zhou [92]		1	✓			✓			Amount left at the re-order point	GA
Gilland and Heese [50]		1	✓							Heuristic algorithm
Jajja [67]		1	✓							System dynamics
Tsao *et al.* [104]	n	1	✓					✓	Promotion level, trade allowances	MINLP, Piecewise linearization technique, Game theory
Castelli and Vanneschi [20]		1	✓							Hybrid GA-VNS
Leng *et al.* [76]		n	✓							Game theory
Wang *et al.* [109]	2	1	✓	✓	✓					Exact solution
Amit *et al.* [2]		1	✓							Stochastic Model, Exact solution
Gajjar and Adil [45]		1	✓							Dynamic programming and Local search
Zhou and Piramuthu [115]		1	✓							Analytical
Geismar *et al.* [48]		1	✓							Heuristic algorithm
Ghazavi and Lotfi [49]		1	✓			✓			Variable demand, Ordering period	Simulation-optimization (GA-ICA)
Flamand *et al.* [40]		1	✓						Aisle/shelf allocation, location of products in shelf	MIP, CPLEX
Zhao *et al.* [112]		1	✓			✓			Shelf display location	SA based hyper heuristic
Kuiti *et al.* [73]	1	1	✓		✓					Exact solution

TABLE 4. continued.

Paper(s)	Supply chain members		Decision variables							Model/solution method
	Manufacturers	Retailers	Space allocated	Wholesale price	Retailer price	Order quantity	Reorder point	Assortment	Others	
Zhao *et al.* [113]	n	1	✓		✓			✓		Exact solution
Beed *et al.* [10]		1	✓		✓			✓		Hyper-heuristic algorithm
Vincent *et al.* [107]		1	✓					✓		Heuristic algorithm
Düsterhöft *et al.* [36]		1	✓				✓	✓		Heuristic algorithm
Kim and Moon [70]		1	✓			✓		✓		Heuristic algorithm
Hübner *et al.* [62]		1	✓			✓		✓		Heuristic algorithm
Ostermeier *et al.* [91]		1	✓					✓		Heuristic algorithm
Czerniachowska *et al.* [29]		1	✓					✓		GA
Sajadi and Ahmadi [99]		1	✓			✓		✓		GA
Czerniachowska *et al.* [30]		1	✓					✓		CPLEX
Gecili and Parikh [47]		1	✓					✓		Hybrid heuristic algorithm
Czerniachowska *et al.* [30]		1	✓					✓		SSAP-NL and SSAP-L
Zheng *et al.* [114]		1	✓							Hybrid GA
Edirisinghe and Munson [37]		n	✓							Association rule mining
Hübner and Kuhn [58]		1	✓				✓	✓		BIP reformulation
Taleizadeh *et al.* [102]	1	1	✓		✓				✓	MINLP, CPLEX
Czerniachowska *et al.* [32]		1	✓					✓		MINLP, CPLEX
Reza [97]		n	✓		✓			✓	✓	Quadratic programming
Chen *et al.* [23]		n	✓							Reformulation linearization technique

enable retailers to use real-time data for more dynamic and adaptive decision-making processes. For example, the application of AI algorithms in predicting consumer behavior can significantly enhance the accuracy of shelf-space allocation beyond traditional modeling techniques. This includes challenges related to data integration from various sources, the need for continuous adaptation to changing consumer preferences, and the implications of e-commerce integration on traditional shelf-space strategies. Hence, examining the collaborative roles of AI and human decision-makers in shelf-space management could provide valuable insights into maximizing efficiency. Furthermore, research could explore the impact of real-time inventory management systems on optimizing shelf space in hybrid retail environments, where both online and offline interactions influence purchasing behavior. These additions will not only enhance the relevance of our study to modern retailing but also serve as a foundation for future research in this rapidly evolving field. All these challenges and concerns can result in new and novel models responding real-world problems, appropriately.

Data availability statement

No new data/codes were created or analyzed in this study.

References

[1] H.K. Alfares, An EPQ model with variable holding cost. *Int. J. Ind. Eng.-Theory App. Pract.* **19** (2012) 232–240.

[2] R. Amit *et al.*, Optimal shelf-space stocking policy using stochastic dominance under supply-driven demand uncertainty. *Eur. J. Oper. Res.* **246** (2015) 339–342.

[3] N. Amrouche and G. Zaccour, Shelf-space allocation of national and private brands. *Eur. J. Oper. Res.* **180** (2007) 648–663.

[4] N. Amrouche and G. Zaccour, A shelf-space-dependent wholesale price when manufacturer and retailer brands compete. *OR Spect.* **31** (2009) 361–383.

[5] E.E. Anderson and H.N. Amato, A mathematical model for simultaneously determining the optimal brand-collection and display-area allocation. *Oper. Res.* **22** (1974) 13–21.

[6] R. Bai and G. Kendall, A model for fresh produce shelf-space allocation and inventory management with freshness-condition-dependent demand. *INFORMS J. Comput.* **20** (2008) 78–85.

[7] R. Bai *et al.*, Heuristic, meta-heuristic and hyper-heuristic approaches for fresh produce inventory control and shelf space allocation. *J. Oper. Res. Soc.* **59** (2008) 1387–1397.

[8] R. Bai *et al.*, A new model and a hyper-heuristic approach for two-dimensional shelf space allocation. *4OR* **11** (2013) 31–55.

[9] O. Baron *et al.*, Shelf space management when demand depends on the inventory level. *Prod. Oper. Manage.* **20** (2011) 714–726.

[10] R.S. Beed *et al.*, A study of the effect of the parameters for optimizing profit using simulated annealing to solve shelf space allocation problem. *Azerbaijan J. High Perform. Comput.* **3** (2020) 255–268.

[11] T. Bianchi-Aguiar *et al.*, Using analytics to enhance a food retailer's shelf-space management. *Interfaces* **46** (2016) 424–444.

[12] J.H. Bookbinder and F.H. Zarour, Direct product profitability and retail shelf-space allocation models. *J. Bus. Logistics* **22** (2001) 183–208.

[13] N. Borin and P. Farris, A sensitivity analysis of retailer shelf management models. *J. Retail.* **71** (1995) 153–171.

[14] N. Borin *et al.*, A model for determining retail product category assortment and shelf space allocation. *Decis. Sci.* **25** (1994) 359–384.

[15] M.G. Brown and J.Y. Lee, Allocation of shelf space: a case study of refrigerated juice products in grocery stores. *Agribusiness* **12** (1996) 113–121.

[16] A. Bultez and P. Naert, SH. ARP: Shelf allocation for retailers' profit. *Marketing Sci.* **7** (1988) 211–231.

[17] A. Bultez *et al.*, Asymmetric cannibalism in retail assortments. *J. Retail.* **65** (1989) 153.

[18] F. Buttle, Retail space allocation. *Int. J. Phys. Distrib. Mater. Manage.* **14** (1984) 3–23.

[19] P. Capik and A. Brockerhoff, The role of strategic partnerships in the internationalisation of small born global firms. *Entrepreneurial Bus. Econ. Rev.* **5** (2017) 49–77.

[20] M. Castelli and L. Vanneschi, Genetic algorithm with variable neighborhood search for the optimal allocation of goods in shop shelves. *Oper. Res. Lett.* **42** (2014) 355–360.

[21] M.-C. Chen and C.-P. Lin, A data mining approach to product assortment and shelf space allocation. *Expert Syst. App.* **32** (2007) 976–986.

[22] Y.-L. Chen *et al.*, A data mining approach for retail knowledge discovery with consideration of the effect of shelf-space adjacency on sales. *Decis. Support Syst.* **42** (2006) 1503–1520.

[23] Y. Chen *et al.*, Omnichannel product selection and shelf space planning optimization. *Omega* **127** (2024) 103074.

[24] J. Chiang and R. Wilcox, A cross-category analysis of shelf-space allocation, product variety, and retail margins. *Marketing Lett.* **8** (1997) 183–191.

[25] D. Cho and Y. Lee, Seasonal supply chain and the bullwhip effect. *Int. J. Ind. Eng.* **20** (2013) 188.

[26] M. Corstjens and P. Doyle, A model for optimizing retail space allocations. *Manage. Sci.* **27** (1981) 822–833.

[27] M. Corstjens and P. Doyle, A dynamic model for strategically allocating retail space. *J. Oper. Res. Soc.* **34** (1983) 943–951.

[28] R.C. Curhan, Shelf space allocation and profit maximization in mass retailing. *J. Marketing* **37** (1973) 54–60.

[29] K. Czerniachowska *et al.*, Genetic algorithm for the retailers' shelf space allocation profit maximization problem. *Appl. Sci.* **11** (2021) 6401.

[30] K. Czerniachowska *et al.*, Linear and nonlinear shelf space allocation problems with vertical and horizontal bands. *J. Econ. Manage.* **44** (2022) 119–141.

[31] K. Czerniachowska *et al.*, Heuristics for dimensioning the shelf space of the rack with vertical and horizontal product categorisation in the distribution centre with zone picking-in progress. *Manage. Prod. Eng. Rev.* **13** (2022).

[32] K. Czerniachowska *et al.*, Industry expertise heuristics for dimensioning shelf space of rack storage location in a distribution centre with zone picking. *Manage. Prod. Eng. Rev.* **14** (2023) 43–60.

[33] Z. Deng and R.R. Sinkovics, Rapid expansion of international new ventures across institutional distance. *J. Int. Bus. Stud.* **49** (2018) 1010–1032.

[34] P. Desmet and V. Renaudin, Estimation of product category sales responsiveness to allocated shelf space. *Int. J. Res. Marketing* **15** (1998) 443–457.

[35] X. Dreze *et al.*, Shelf management and space elasticity. *J. Retail.* **70** (1994) 301–326.

[36] T. Düsterhöft *et al.*, A practical approach to the shelf-space allocation and replenishment problem with heterogeneously sized shelves. *Eur. J. Oper. Res.* **282** (2020) 252–266.

[37] G.S. Edirisinghe and C.L. Munson, Strategic rearrangement of retail shelf space allocations: using data insights to encourage impulse buying. *Expert Syst. App.* **216** (2023) 119442.

[38] M. Eisend, Shelf space elasticity: a meta-analysis. *J. Retail.* **90** (2014) 168–181.

[39] M.M. Fadıloğlu *et al.*, A model and case study for efficient shelf usage and assortment analysis. *Ann. Oper. Res.* **180** (2010) 105–124.

[40] T. Flamand *et al.*, Promoting impulse buying by allocating retail shelf space to grouped product categories. *J. Oper. Res. Soc.* **67** (2016) 953–969.

[41] T. Flamand *et al.*, Integrated assortment planning and store-wide shelf space allocation: an optimization-based approach. *Omega* **81** (2018) 134–149.

[42] H.K. Gajjar and G.K. Adil, A piecewise linearization for retail shelf space allocation problem and a local search heuristic. *Ann. Oper. Res.* **179** (2010) 149–167.

[43] H.K. Gajjar and G.K. Adil, A dynamic programming heuristic for retail shelf space allocation problem. *Asia-Pac. J. Oper. Res.* **28** (2011) 183–199.

[44] H.K. Gajjar and G.K. Adil, Heuristics for retail shelf space allocation problem with linear profit function. *Int. J. Retail Distrib. Manage.* **39** (2011) 144–155.

[45] H.K. Gajjar and G.K. Adil, Retail shelf space allocation considering inventory replenishment. *Int. J. Serv. Oper. Manage.* **22** (2015) 221–234.

[46] J. Gao and Z. Meng, Joint decision model of variants selection shelf-space allocation and inventory control. *J. Syst. Eng.* **5** (2009) 615–620.

[47] H. Gecili and P.J. Parikh, Joint shelf design and shelf space allocation problem for retailers. *Omega* **111** (2022) 102634.

[48] H.N. Geismar *et al.*, Maximizing revenue through two-dimensional shelf-space allocation. *Prod. Oper. Manage.* **24** (2015) 1148–1163.

[49] E. Ghazavi and M. Lotfi, Formulation of customers' shopping path in shelf space planning: a simulation-optimization approach. *Expert Syst. App.* **55** (2016) 243–254.

[50] W.G. Gilland and H.S. Heese, Sequence matters: shelf-space allocation under dynamic customer-driven substitution. *Prod. Oper. Manage.* **22** (2013) 875–887.

[51] E. Grigoroudis and M. Doumpos, Operational Research in Business and Economics. Springer (2017).

[52] P. Hansen and H. Heinsbroek, Product selection and space allocation in supermarkets. *Eur. J. Oper. Res.* **3** (1979) 474–484.

[53] J.M. Hansen *et al.*, Retail shelf allocation: a comparative analysis of heuristic and meta-heuristic approaches. *J. Retail.* **86** (2010) 94–105.

[54] M.A. Hariga and A. Al-Ahmari, An integrated retail space allocation and lot sizing models under vendor managed inventory and consignment stock arrangements. *Comput. Ind. Eng.* **64** (2013) 45–55.

[55] M.A. Hariga *et al.*, A joint optimisation model for inventory replenishment, product assortment, shelf space and display area allocation decisions. *Eur. J. Oper. Res.* **181** (2007) 239–251.

[56] A. Hübner and H. Kuhn, Retail shelf space management model with space-elastic demand and consumer–driven substitution effects (2011).

[57] A.H. Hübner and H. Kuhn, Retail category management: State-of-the-art review of quantitative research and software applications in assortment and shelf space management. *Omega* **40** (2012) 199–209.

[58] A. Hübner and H. Kuhn, Decision support for managing assortments, shelf space, and replenishment in retail. *Flexible Serv. Manuf. J.* **36** (2023) 1–35.

[59] A. Hübner and K. Schaal, Effect of replenishment and backroom on retail shelf-space planning. *Bus. Res.* **10** (2017) 123–156.

[60] A. Hübner and K. Schaal, An integrated assortment and shelf-space optimization model with demand substitution and space-elasticity effects. *Eur. J. Oper. Res.* **261** (2017) 302–316.

[61] A. Hübner *et al.*, Maximizing profit via assortment and shelf-space optimization for two-dimensional shelves. *Prod. Oper. Manage.* **29** (2020) 547–570.

[62] A. Hübner *et al.*, Shelf space dimensioning and product allocation in retail stores. *Eur. J. Oper. Res.* **292** (2021) 155–171.

[63] H. Hwang *et al.*, A model for shelf space allocation and inventory control considering location and inventory level effects on demand. *Int. J. Prod. Econ.* **97** (2005) 185–195.

[64] H. Hwang *et al.*, A genetic algorithm approach to an integrated problem of shelf space design and item allocation. *Comput. Ind. Eng.* **56** (2009) 809–820.

[65] J. Irion *et al.*, A hierarchical decomposition approach to retail shelf space management and assortment decisions. *J. Oper. Res. Soc.* **62** (2011) 1861–1870.

[66] J. Irion *et al.*, A piecewise linearization framework for retail shelf space management models. *Eur. J. Oper. Res.* **222** (2012) 122–136.

[67] M. Jajja, Retail shelf space allocation analysis using system dynamics approach. *J. Qual. Technol. Manage.* **IX** (2013) 99–129.

[68] A.C.R. Janeiro, *Optimization algorithms for the shelf space allocation problem.* Master's thesis, Universidade do Porto (Portugal) (2014).

[69] P. Kanjalkar *et al.*, Intelligent inventory management in retail stores using four state of the art object detection models, in International Conference on Information Science and Applications. Springer (2023).

[70] G. Kim and I. Moon, Integrated planning for product selection, shelf-space allocation, and replenishment decision with elasticity and positioning effects. *J. Retail. Consum. Serv.* **58** (2021) 102274.

[71] M.A. Koschat, Store inventory can affect demand: empirical evidence from magazine retailing. *J. Retail.* **84** (2008) 165–179.

[72] J.A. Kotzan and R.V. Evanson, Responsiveness of drug store sales to shelf space allocations. *J. Market. Res.* **6** (1969) 465–469.

[73] M.R. Kuiti *et al.*, Integrated product design, shelf-space allocation and transportation decisions in green supply chains. *Int. J. Prod. Res.* **57** (2019) 6181–6201.

[74] C.-W. Kuo and S.-J. S. Yang, The role of store brand positioning for appropriating supply chain profit under shelf space allocation. *Eur. J. Oper. Res.* **231** (2013) 88–97.

[75] M. Leng *et al.*, The retail space-exchange problem with pricing and space allocation decisions. *Prod. Oper. Manage.* **22** (2013) 189–202.

[76] M. Leng *et al.*, Cooperative game analysis of retail space-exchange problems. *Eur. J. Oper. Res.* **232** (2014) 393–404.

[77] X. Li *et al.*, Game theory methodology for optimizing retailers' pricing and shelf-space allocation decisions on competing substitutable products. *Int. J. Adv. Manuf. Technol.* **68** (2013) 375–389.

[78] A. Lim *et al.*, Metaheuristics with local search techniques for retail shelf-space optimization. *Manage. Sci.* **50** (2004) 117–131.

[79] M. Lotfi and S.A. Torabi, A fuzzy goal programming approach for mid-term assortment planning in supermarkets. *Eur. J. Oper. Res.* **213** (2011) 430–441.

[80] M.M. Lotfi *et al.*, A weighted goal programming approach for replenishment planning and space allocation in a supermarket. *J. Oper. Res. Soc.* **62** (2011) 1128–1137.

[81] M.K. Maiti and M. Maiti, Multi-item shelf-space allocation of breakable items via genetic algorithm. *J. Appl. Math. Comput.* **20** (2006) 327–343.

[82] G. Martín-Herrán and S. Taboubi, Shelf-space allocation and advertising decisions in the marketing channel: a differential game approach. *Int. Game Theory Rev.* **7** (2005) 313–330.

[83] G. Martín-Herrän and S. Taboubi, Incentive strategies for shelf-space allocation in duopolies, in Dynamic Games: Theory and Applications. Springer (2005) 231–253.

[84] G. Martín-Herrán *et al.*, The impact of manufacturers' wholesale prices on a retailer's shelf-space and pricing decisions. *Decis. Sci.* **37** (2006) 71–90.

[85] V. Martínez-de-Albéniz and G. Roels, Competing for shelf space. *Prod. Oper. Manage.* **20** (2011) 32–46.

[86] I. Moussaoui *et al.*, Drivers of retail on-shelf availability: systematic review, critical assessment, and reflections on the road ahead. *Int. J. Phys. Distrib. Logistics Manage.* **46** (2016) 516–535.

[87] C.C. Murray *et al.*, Joint optimization of product price, display orientation and shelf-space allocation in retail category management. *J. Retail.* **86** (2010) 125–136.

[88] M. Nafari and J. Shahrabi, A temporal data mining approach for shelf-space allocation with consideration of product price. *Expert Syst. App.* **37** (2010) 4066–4072.

[89] K. Nakade and K. Ikeuchi, Optimal ordering and pricing on clearance goods. *Int. J. Ind. Eng.* **23** (2016) 155.

[90] A.F. Nogales and M.G. Suarez, Shelf space management of private labels: a case study in Spanish retailing. *J. Retail. Consum. Serv.* **12** (2005) 205–216.

[91] M. Ostermeier *et al.*, A model and solution approach for store-wide shelf space allocation. *Omega* **102** (2021) 102425.

[92] S. Piramuthu and W. Zhou, RFID and perishable inventory management with shelf-space and freshness dependent demand. *Int. J. Prod. Econ.* **144** (2013) 635–640.

[93] M. Rabbani *et al.*, Profit maximizing through 3D shelf space allocation of 2D display orientation items with variable heights of the shelves. *Opsearch* **55** (2018) 337–360.

[94] B. Ramaseshan *et al.*, Decision support tool for retail shelf space optimization. *Int. J. Inf. Technol. Decis. Making* **7** (2008) 547–565.

[95] S. Raut *et al.*, Heuristic and meta-heuristic approaches for multi-period shelf-space optimization: the case of motion picture retailing. *J. Oper. Res. Soc.* **60** (2009) 1335–1348.

[96] P.M. Reyes and G.V. Frazier, Goal programming model for grocery shelf space allocation. *Eur. J. Oper. Res.* **181** (2007) 634–644.

[97] M. Reza, Stackelberg game theory model with price and shelf space dependent demand under revenue sharing contract. *JIMS* **21** (2023) 1–34.

[98] R.A. Russell and T.L. Urban, The location and allocation of products and product families on retail shelves. *Ann. Oper. Res.* **179** (2010) 131–147.

[99] S.J. Sajadi and A. Ahmadi, An integrated optimization model and metaheuristics for assortment planning, shelf space allocation, and inventory management of perishable products: a real application. *PloS One* **17** (2022) e0264186.

[100] M.S. Sajadieh and M. Danaei, Robust optimization approach for pricing and shelf space decisions with uncertain demand. *Sci. Iran.* **29** (2022) 303–319.

[101] K. Schaal and A. Hübner, When does cross-space elasticity matter in shelf-space planning? A decision analytics approach. *Omega* **80** (2018) 135–152.

[102] A.A. Taleizadeh *et al.*, Pricing, warranty, and shelf space decisions for the supply chain with non-symmetric market and warranty-period dependent demand. *Oper. Res.* **23** (2023) 61.

[103] C.-Y. Tsai and S.-H. Huang, A data mining approach to optimise shelf space allocation in consideration of customer purchase and moving behaviours. *Int. J. Prod. Res.* **53** (2015) 850–866.

[104] Y.-C. Tsao *et al.*, Retailer shelf-space management with trade allowance: a Stackelberg game between retailer and manufacturers. *Int. J. Prod. Econ.* **148** (2014) 133–144.

[105] T.L. Urban, An inventory-theoretic approach to product assortment and shelf-space allocation. *J. Retail.* **74** (1998) 15–35.

[106] F.Y. Vincent *et al.*, Shelf space allocation problem under carbon tax and emission trading policies. *J. Clean. Prod.* **196** (2018) 438–451.

[107] F.Y. Vincent *et al.* A reduced variable neighborhood search-based hyperheuristic for the shelf space allocation problem. *Comput. Ind. Eng.* **143** (2020) 106420.

[108] Y. Wang and Y. Gerchak, Supply chain coordination when demand is shelf-space dependent. *Manuf. Serv. Oper. Manage.* **3** (2001) 82–87.

[109] S.-Y. Wang *et al.*, Pricing and shelf space decisions with non-symmetric market demand. *Int. J. Prod. Econ.* **169** (2015) 233–239.

[110] M.-H. Yang, An efficient algorithm to allocate shelf space. *Eur. J. Oper. Res.* **131** (2001) 107–118.

[111] M.-H. Yang and W.-C. Chen, A study on shelf space allocation and management. *Int. J. Prod. Econ.* **60** (1999) 309–317.

[112] J. Zhao *et al.*, Joint optimization models for shelf display and inventory control considering the impact of spatial relationship on demand. *Eur. J. Oper. Res.* **255** (2016) 797–808.

[113] J. Zhao *et al.*, The shelf space and pricing strategies for a retailer-dominated supply chain with consignment based revenue sharing contracts. *Eur. J. Oper. Res.* **280** (2020) 926–939.

[114] L. Zheng *et al.*, A data-driven model assisted hybrid genetic algorithm for a two-dimensional shelf space allocation problem. *Swarm Evol. Comput.* **77** (2023) 101251.

[115] W. Zhou and S. Piramuthu, Effect of ticket-switching on inventory and shelf-space allocation. *Decis. Support Syst.* **69** (2015) 31–39.

[116] M. Ziari, A data-driven pricing model for distribution systems considering competition. *Adv. Ind. Eng.* **58** (2024) 179–195.

[117] M. Ziari and M.S. Sajadieh, Behavior-based pricing considering competition in retail systems. *Sharif J. Ind. Eng. Manage.* **38** (2023) 59–66.

[118] F.S. Zufryden, A dynamic programming approach for product selection and supermarket shelf-space allocation. *J. Oper. Res. Soc.* **37** (1986) 413–422.

[119] F.S. Zufryden, New computational results with a dynamic programming approach for product selection and supermarket shelf-space allocation. *J. Oper. Res. Soc.* **38** (1987) 201–203.

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